

Lecture 7 Continue

{ Pre-integration continued
 Optimization based VINS (graph VINS)
 Multi-visual-inertial system

we start with the abstracted IMU function

From the time k to time j ,

$${}^G_{I_j}R = {}^G_{I_k}R \cdot \Delta R$$

$${}^G_{P_j} = {}^G_{P_k} + {}^G_{V_k} \Delta t + {}^G_{I_k}R \Delta P + \frac{1}{2} {}^G_g \Delta t^2 \quad \Delta t = t_j - t_k$$

$${}^G_{V_j} = {}^G_{V_k} + {}^G_{I_k}R \Delta V + {}^G_g \Delta t$$

$$b_{gj} = b_{gk} + \int_{t_k}^{t_j} n_{wg} dt$$

$$b_{aj} = b_{ak} + \int_{t_k}^{t_j} n_{wa} dt$$

Then $\Delta R, \Delta P, \Delta V$ are the integration terms, $\begin{cases} z_a = z_{a_k} - b_{a_k} - n_a \\ z_w = z_{w_k} - b_{g_k} - n_g \end{cases}$

$$\Delta R(b_g, n_z) = {}^G_{I_k}R^T {}^G_{I_j}R$$

$$\Delta P(b_g, b_a, n_z) = {}^G_{I_k}R^T ({}^G_{P_j} - {}^G_{P_k} - {}^G_{V_k} \Delta t - \frac{1}{2} {}^G_g \Delta t^2)$$

$$\Delta V(b_g, b_a, n_z) = {}^G_{I_k}R^T ({}^G_{V_j} - {}^G_{V_k} - {}^G_g \Delta t)$$

$$\Delta b_g = b_{gj} - b_{gk}$$

$$\Delta b_a = b_{aj} - b_{ak}$$

$$\Rightarrow \begin{cases} \Delta R(b_g, n_z) \triangleq h_r(x_k, x_j) \\ \Delta P(b_g, b_a, n_z) \triangleq h_p(x_k, x_j) \\ \Delta V(b_g, b_a, n_z) \triangleq h_v(x_k, x_j) \\ \Delta b_g = b_{gj} - b_{gk} \\ \Delta b_a = b_{aj} - b_{ak} \end{cases}$$

As we discussed, when we use the above cost in the least squares,

$\Delta R, \Delta P$ and ΔV needs to be re-integrated for every iterations in Gaussian Newton

or LM algorithms.

Hence: what's the way to only integrate once and avoid repeated integration?

$$z = h(x_A, x_B) + n$$

$$\tilde{z} = H_A \tilde{x}_A + H_B^0 \tilde{x}_B^0 + n$$

$$\tilde{x}_B^0 = x_B - \hat{x}_B^0 = x_B - \hat{x}_B + \hat{x}_B - \hat{x}_B^0 = \tilde{x}_B + \Delta \hat{x}_B$$

$$\Rightarrow \tilde{x}_B^0 = \tilde{x}_B + \Delta \hat{x}_B$$

$$= H_A \tilde{x}_A + H_B^0 \tilde{x}_B + H_B^0 \Delta \hat{x}_B + n$$

$$\Rightarrow \tilde{z} - \boxed{H_B^0 \Delta \hat{x}_B} = \underline{H_A \tilde{x}_A} + \underline{H_B^0 \tilde{x}_B} + n$$

H_B^0 : linearize at $\hat{x}_B^0$, which is the initial estimate.

Hence, apply the above:

$$\Delta R(b_g, n_2) = \Delta R(\hat{b}_g^0 + \tilde{b}_g^0, n_2) = \Delta R(\hat{b}_g^0, 0) \cdot \exp(H_g^R \tilde{b}_g^0 + H_n^R n_2)$$

$$= \Delta R(\hat{b}_g^0, 0) \cdot \exp(H_g^R \tilde{b}_g^0 + H_n^R n_2)$$

$$= \Delta R(\hat{b}_g^0, 0) \cdot \exp(H_g^R \Delta \hat{x}_{b_g} + H_g^R \tilde{b}_g + H_n^R n_2) \quad \Delta \hat{x}_{b_g} = \hat{b}_g - \hat{b}_g^0$$

$$= h_R(x_k, x_j)$$

$$\Delta R(\hat{b}_g^0, 0) = h_R(x_k, x_j) \cdot \exp(-H_g^R \Delta \hat{x}_{b_g} - H_g^R \tilde{b}_g - H_n^R n_2)$$

$$\Delta p(b_g, b_a, n_2) = \Delta p(\hat{b}_g^0, \hat{b}_a^0, 0) + H_b^P \Delta \hat{x}_b + H_b^P \tilde{x}_b + H_n^P n_2$$

$$= h_p(x_k, x_j)$$

$$x_b = \begin{bmatrix} b_g \\ b_a \end{bmatrix}$$

$$\Delta x_b = \begin{bmatrix} \hat{b}_g - \hat{b}_g^0 \\ \hat{b}_a - \hat{b}_a^0 \end{bmatrix}$$

$$\Delta p(\hat{b}_g^0, \hat{b}_a^0, 0) = h_p(x_k, x_j) - H_b^P \Delta \hat{x}_b - H_b^P \tilde{x}_b - H_n^P n_2$$

$$\Delta v(b_g, b_a, n_2) = \Delta v(\hat{b}_g^0, \hat{b}_a^0, 0) + H_b^V \Delta \hat{x}_b + H_b^V \tilde{x}_b + H_n^V n_2$$

$$= h_v(x_k, x_j)$$

$$\Delta v(\hat{b}_g^0, \hat{b}_a^0, 0) = h_v(x_k, x_j) - H_b^V \Delta \hat{x}_b - H_b^V \tilde{x}_b - H_n^V n_2$$

Hence: we finally have the IMU measurements:

$$\begin{bmatrix} \Delta R(\hat{b}_g^0, 0) \\ \Delta p(\hat{b}_g^0, \hat{b}_a^0, 0) \\ \Delta v(\hat{b}_g^0, \hat{b}_a^0, 0) \\ \Delta b_g \\ \Delta b_a \end{bmatrix} = \begin{bmatrix} h_R(x_k, x_j) \cdot \text{Exp}(-H_g^R \Delta \hat{x}_{bg} - H_g^R \tilde{b}_g - H_n^R n_z) \\ h_p(x_k, x_j) - H_b^p \Delta \hat{x}_b - H_b^p \tilde{x}_b - H_n^p n_z \\ h_v(x_k, x_j) - H_b^v \Delta \hat{x}_b - H_b^v \tilde{x}_b - H_n^v n_z \\ b_{gj} - b_{gk} \\ b_{aj} - b_{ak} \end{bmatrix}$$

$$z_1 = h_1(x_k, x_j, n_z)$$

z_1 : we only need to integrate once given $\hat{b}_g^0, \hat{b}_a^0$

$\Delta x_b = \begin{bmatrix} \hat{b}_g - \hat{b}_g^0 \\ \hat{b}_a - \hat{b}_a^0 \end{bmatrix}$ is the bias linearization change correction:

now: we can construct IMU factor or IMU cost term:

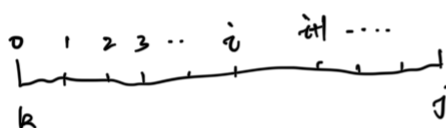
$$G_1 = \|z_1 - h_1(x_k, x_j, 0)\|_{Q_1}^2 \quad n_z \sim \mathcal{N}(0, Q_1)$$

z_1 : is IMU pre-integrated measurements

n_z : is IMU pre-integrated noise, with Q_1 as covariance

now: How to do the IMU pre-integration and get:

$$z_1, Q_1, H_b^p, H_b^v, H_{bg}^R$$



IMU readings between k and j .
 $i \in [k, j]$

The way to solve it is to use recursive formulation:

$$\begin{cases} \Delta R_{i,t+1} = \Delta R_i \cdot R_{i,t+1} \\ \Delta P_{i,t+1} = \Delta P_i + \Delta V_i \cdot \Delta t_i + \Delta R_i \cdot P_{i,t+1} \\ \Delta V_{i,t+1} = \Delta V_i + \Delta R_i \cdot V_{i,t+1} \\ \Delta b_{gi,t+1} = \Delta b_{gi} + \int_{t_{k,i}}^{t_{k+1,i}} n_{wg} dt \\ \Delta b_{ai,t+1} = \Delta b_{ai} + \int_{t_{k,i}}^{t_{k+1,i}} n_{wa} dt \end{cases}$$

$$Z_i \rightarrow Z_{i,t+1}$$

If we do linearization:

$$\tilde{Z}_{i,t+1} = \Phi_{(i,t+1,i)} \tilde{Z}_{i,t} + \bar{\Phi}_b \tilde{x}_{b,t} + G_i n_{2d}$$

$$Q_{i,t+1} = \Phi_{(i,t+1,i)} Q_{i,t} \Phi_{(i,t+1,i)}^T + G_i Q_d G_i^T$$

$\bar{\Phi}_b$: contains the Jacobians for: H_g^R, H_b^P, H_b^V

$$R_{i,t+1} = \text{Exp}(\hat{\omega}_i \Delta t)$$

$$P_{i,t+1} = \int_{t_{k,i}}^{t_{k+1,i}} \int_S \frac{I_{k,i}}{I_g R} \frac{I_g}{a} ds d\tau$$

$$V_{i,t+1} = \int_{t_{k,i}}^{t_{k+1,i}} \frac{I_{k,i}}{I_g} R \cdot \frac{I_g}{a} d\tau$$

use AC7, CP1, linear assumption to do the integration:

use RK4 to do the integration.

n_{2d} : discrete IMU noise

$$n_{2d} \sim \mathcal{N}\left(0, \begin{bmatrix} \sigma_g^2 I_3 & & & \\ & \sigma_a^2 I_3 & & \\ & & \sigma_{\omega_g}^2 I_3 & \\ & & & \sigma_{w_a}^2 I_2 \end{bmatrix} \frac{1}{\Delta t}\right)$$

now: it is complete for the IMU pre-integration factor:

Discussion

① The key idea: control the linearization of biases, the assumption is that biases is slow varying

② This pre-integration idea can be applied to any other temporal signal:
wheel encoder
rotor rpm

③ This pre-integration keep linearization of biases, but it can be extended to more parameter:

IMU intrinsic: scale/slew parameters

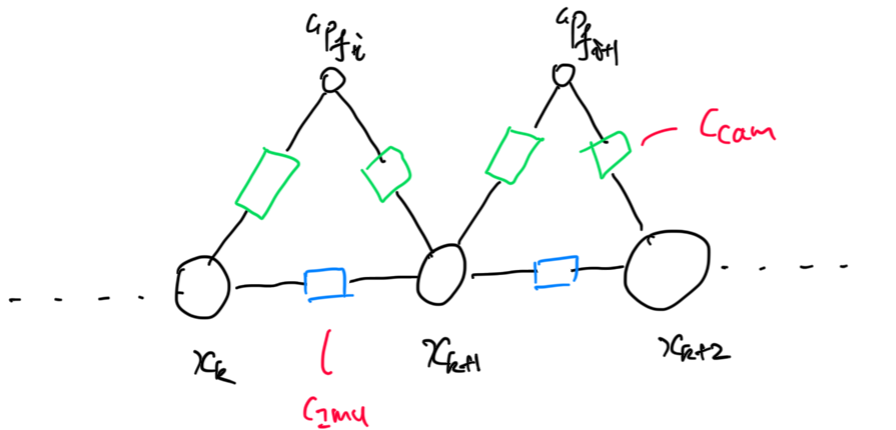
④ How to use IMU factor?

initialize the velocity and biases

A typical optimization based VINS:

$$\begin{cases} x_k \sim \mathcal{N}(\hat{x}_k, P_k) \\ \dot{x} = f(x, u) \rightarrow x_{k+1} = f_{int}(x_k, u_k) \Rightarrow z_1 = h(x_k, x_{k+1}, u) \\ z_{k+1} = h(x_k) + n_{k+1} \end{cases}$$

$$C_{VINS} = \underbrace{\|x_k - \hat{x}_k\|_{P_k}^2}_{C_{IMU}} + \underbrace{\|z_1 - h(x_k, x_{k+1}, u)\|_{G \otimes G^T}^2}_{C_{cam}} + \underbrace{\|z_{k+1} - h(x_{k+1})\|_{P_{k+1}}^2}_{C_{cam}}$$



If purely C_{cam} : $\rightarrow$ BA

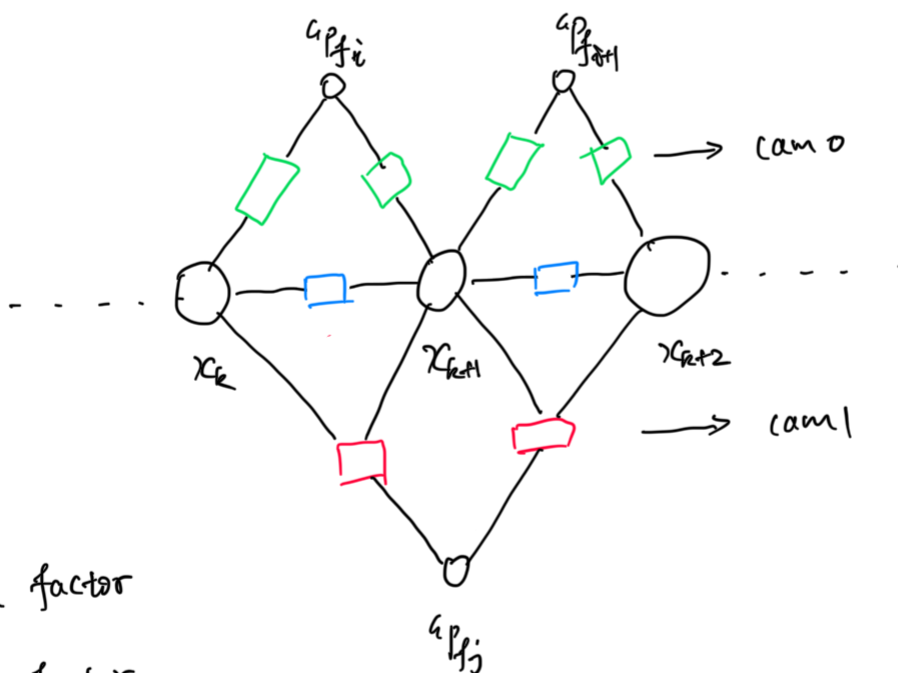
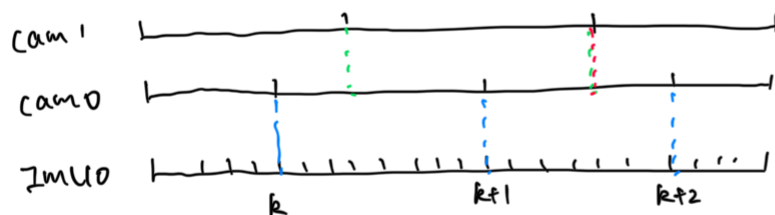
If $C_{IMU} + C_{cam}$: $\rightarrow$ Visual-inertial BA: $\begin{cases} \text{velocity} \\ \text{biases} \end{cases}$ are added to the states

multi-visual-inertial system

what if we have two asynchronous cameras:

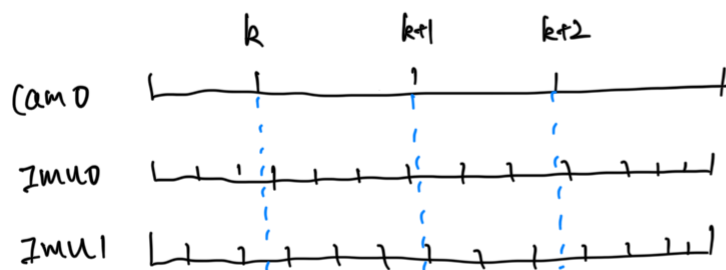
$$G_{C1}^T = G_{I_{k+1}}^T \cdot \underline{I_{C1}^T}$$

$G_{I_{k+1}}^T$ is pose interpolation between $G_{I_{k+1}}^T$ and $G_{I_{k+2}}^T$



- : IMU factor
- : cam factor
- : Interpolated camera factor

what if we have multiple IMUs?



method 1: transfer the IMU readings from IMU I

$${}^a p_{2,1} = {}^a p_{2,0} + {}^{I_0}R^{a,0} r_{2,1}$$

$$\dot{{}^a p}_{2,1} = \dot{{}^a p}_{2,0} + \dot{{}^{I_0}R} {}^{I_0}p_{2,1}$$

$${}^a v_{2,1} = {}^a v_{2,0} + \dot{{}^{I_0}R} \cdot L^{I_0} \omega_{I_0} {}^{I_0}p_{2,1}$$

$$\dot{{}^a v}_{2,1} = \dot{{}^a v}_{2,0} + \dot{{}^{I_0}R} \cdot L^{I_0} \omega_{I_0} {}^{I_0}p_{2,1} + \dot{{}^{I_0}R} \cdot L^{I_0} \dot{\omega}_{I_0} {}^{I_0}p_{2,1}$$

$${}^a a_{2,1} = {}^a a_{2,0} + \dot{{}^{I_0}R} L^{I_0} \omega_{I_0}^2 {}^{I_0}p_{2,1} + \dot{{}^{I_0}R} L^{I_0} \dot{\omega}_{I_0} {}^{I_0}p_{2,1}$$

$${}^{I_1}a_{2,1} = \frac{{}^{I_1}r_{2,1}}{{}^{I_0}R} \cdot ({}^{I_0}a_{2,0} + L^{I_0} \omega_{I_0}^2 {}^{I_0}p_{2,1} + \underbrace{L^{I_0} \dot{\omega}_{I_0} {}^{I_0}p_{2,1}}_{\text{angular acceleration is needed to do the transfer}})$$

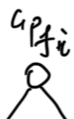
angular acceleration is needed to do the transfer.

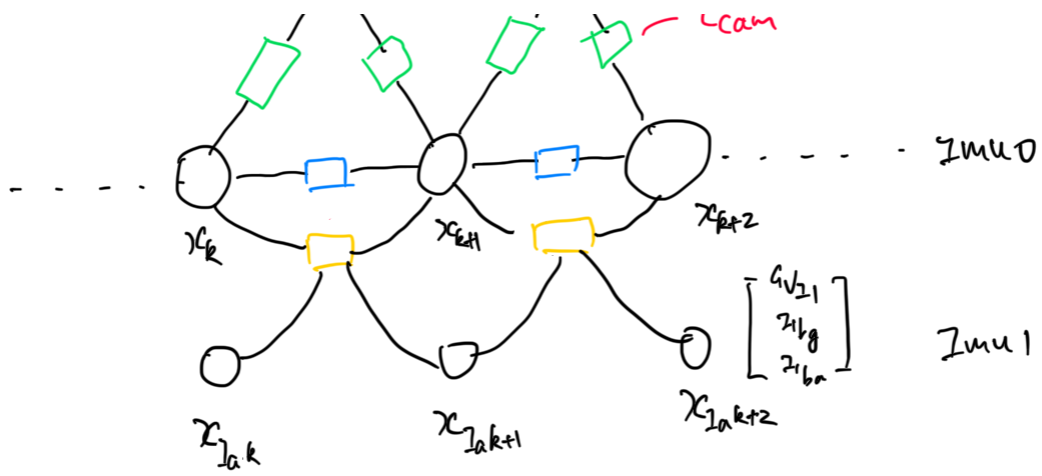
method 2:

$$\begin{cases} {}^a_{I_1}T = {}^a_{I_0}T \cdot {}^{I_0}_{I_1}T \\ {}^a v_{2,1} = {}^a v_{2,0} + \dot{{}^{I_0}R} L^{I_0} \omega_{I_0} {}^{I_0}p_{2,1} \end{cases}$$

Based on the rigid transform: we can represent X_{2mu1} with X_{2mu0}

$$X_{2mu0} = \begin{bmatrix} {}^a_{I_0}R \\ {}^a p_{2,0} \\ {}^a v_{2,0} \\ {}^{I_0}bg \\ {}^{I_0}ba \end{bmatrix} \quad X_{2mu1} = \begin{bmatrix} {}^a_{I_1}R \\ {}^a p_{2,1} \\ {}^a v_{2,1} \\ {}^{I_1}bg \\ {}^{I_1}ba \end{bmatrix}$$





- : base Imu factor: $Imu0$
- : Auxillary Imu factor: $Imu1$
- : camera factor: $cam0$